

Methodology

1. Data Acquisition

- Satellite data were acquired from the Google Earth Engine cloud platform.
- Landsat 8 Collection 2 Level-2 imagery was used for the analysis.
- The study period was selected based on cloud-free conditions.

2. Study Area Preparation

- The study area boundary (AOI) was imported into Google Earth Engine.
- All satellite images were clipped to the study area extent.

3. Image Preprocessing

The preprocessing steps included:

- Cloud and cloud shadow masking using the QA_PIXEL band.
- Application of radiometric scaling factors.
- Creation of a median composite image to reduce atmospheric noise.

4. NDVI Calculation

The Normalized Difference Vegetation Index (NDVI) was calculated using:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

where:

- NIR = Band 5
- Red = Band 4

The NDVI values were used to estimate vegetation abundance.

5. Fractional Vegetation (FV)

Fractional vegetation was calculated as:

$$FV = \left(\frac{NDVI - NDVI_{\min}}{NDVI_{\max} - NDVI_{\min}} \right)^2$$

6. Surface Emissivity (EM)

Surface emissivity was estimated using:

$$EM = 0.004 \times FV + 0.986$$

7. Land Surface Temperature (LST)

Land Surface Temperature was calculated using the thermal infrared band (Band 10):

$$LST = \frac{TB}{1 + \left(\frac{\lambda}{14380} \right) \ln(EM)}$$

where:

- TB = Brightness Temperature (Kelvin)
- λ = Wavelength of Landsat 8 thermal band
- EM = Surface emissivity

The LST values were converted from Kelvin to degrees Celsius.

8. Water Body Removal

The Normalized Difference Water Index (NDWI) was calculated:

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

Water bodies were masked out to improve the accuracy of urban heat island analysis.

9. Urban Heat Island (UHI) Analysis

The Urban Heat Island intensity was calculated using the Z-score normalization approach:

$$UHI = \frac{LST - LST_{mean}}{LST_{std}}$$

where:

- LST_{mean} = Mean land surface temperature
- LST_{std} = Standard deviation of land surface temperature

10. Visualization and Export

The final outputs included:

- True Color Composite

- NDVI Map
- Land Surface Temperature (LST) Map
- Water Mask
- Urban Heat Island (UHI) Map
- Statistical summaries